

vLLM-Omni

A framework for efficient model inference with omni-modality models

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Part 01

Overview



Our Goal



Build the **fastest** and
easiest-to-use open-source
Omni-Modality model
inference & serving engine

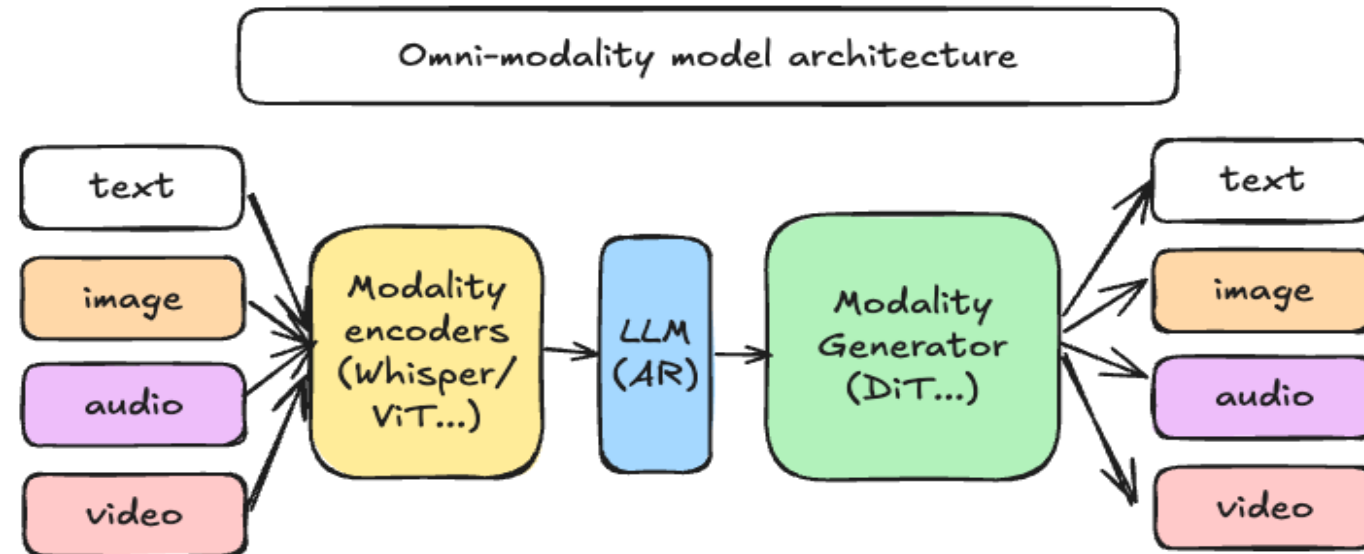


Omni-Modality models

Omni-modality: Text, image, video, and audio data processing

Non-autoregressive Architectures: extend the AR support of vLLM to Diffusion Transformers (DiT) and other parallel generation models

Heterogeneous outputs: from traditional text generation to multimodal outputs



vLLM-Omni API (1): Omni class

A Python interface for offline batched inference for Qwen3-Omni/Qwen-Image

```
from vllm_omni import Omni
```

Example prompts.

```
inputs = {"prompt": prompt,  
         "multi_modal_data":  
         {"video": video_frames,  
          "audio": audio_data},  
         }
```

Create an omni with HF model name.

```
omni = Omni(model="Qwen/Qwen3-  
Omni-30B-A3B-Instruct")
```

Generate texts and audio from the multi-modality inputs.

```
outputs = omni.generate(inputs)
```

```
from vllm_omni import Omni
```

Example prompts.

```
inputs = "A cup of coffee on the table"
```

Create an omni with HF model name.

```
omni = Omni(model="Qwen/Qwen-  
Image-2512")
```

Generate image from multi-modality inputs.

```
outputs = omni.generate(inputs)
```



vLLM-Omni API (2): OpenAI-compatible server



vLLM-Omni

A FastAPI-based server for online serving for Qwen3-Omni

Server

```
$ vllm serve Qwen/Qwen3-Omni-30B-A3B-Instruct --omni --port 8091
```

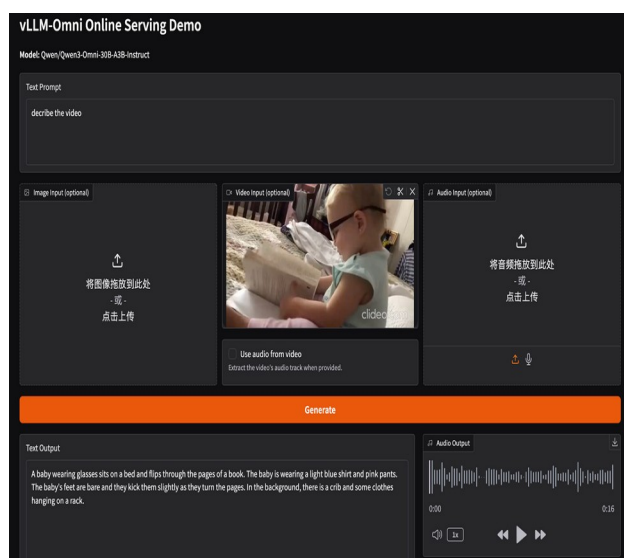
Client

```
$ curl -sS -X POST http://localhost:8091/v1/chat/completions\  
-H "Content-Type: application/json" \  
-d '{  
  "model": "Qwen/Qwen3-Omni-30B-A3B-Instruct",  
  "messages": "Why is this video funny? "  
  "sampling_params_list": $sampling_params_list,  
}'
```



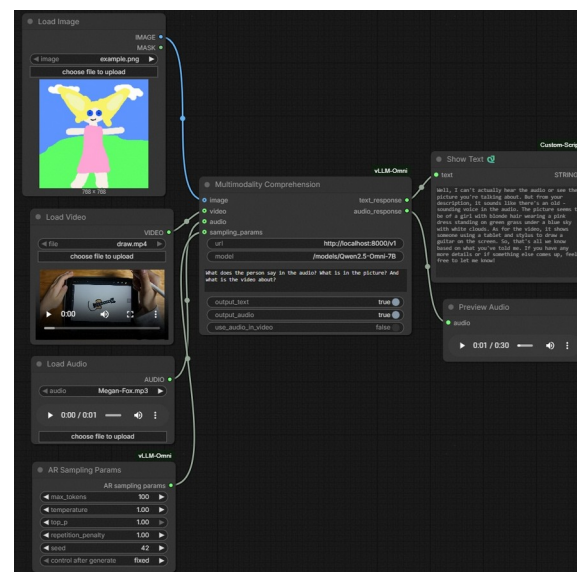
vLLM-Omni serving demo

A Gradio demo for Qwen3-Omni online serving



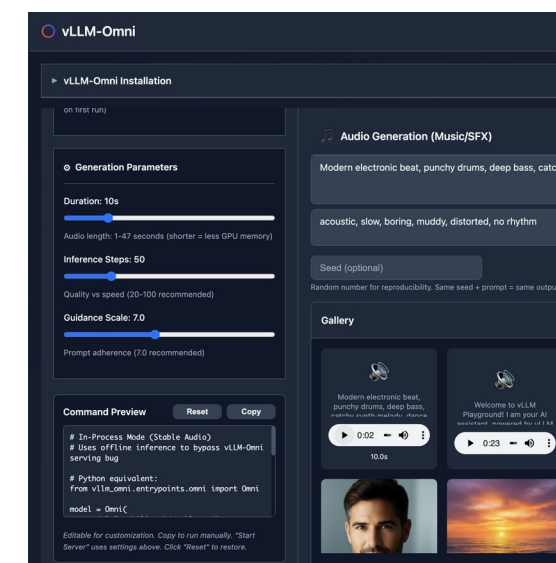
https://docs.vllm.ai/projects/vllm-omni/en/latest/user_guide/examples

ComfyUI Integration for low-code development



<https://docs.vllm.ai/projects/vllm-omni/en/latest/features/comfyui/>

vllm-playground: A web interface for interacting with vLLM servers



<https://github.com/micytao/vllm-playground>



Broad Model Support

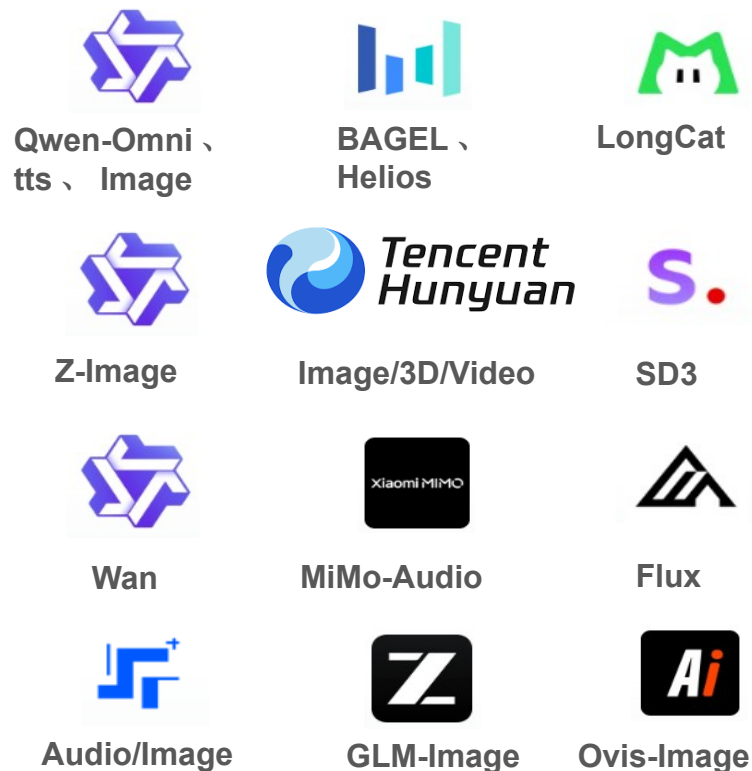


- vLLM-Omni supports 30+ popular omni and diffusion model architectures (growing rapidly)



Qwen @Alibaba_Qwen · Jan 2
Big thanks to @vllm_project for day-zero support of Qwen-Image-2512! 🎉

vLLM @vllm_project · Dec 31, 2025
Congrats to @Alibaba_Qwen on the release of Qwen-Image-2512! 🎉
We are thrilled to announce Day-0 support in vLLM-Omni. You can now serve this SOTA open-source Image model with our optimized pipelined architecture immediately.



...plus **Many** more



Github Repo



<https://github.com/vllm-project/vllm-omni>



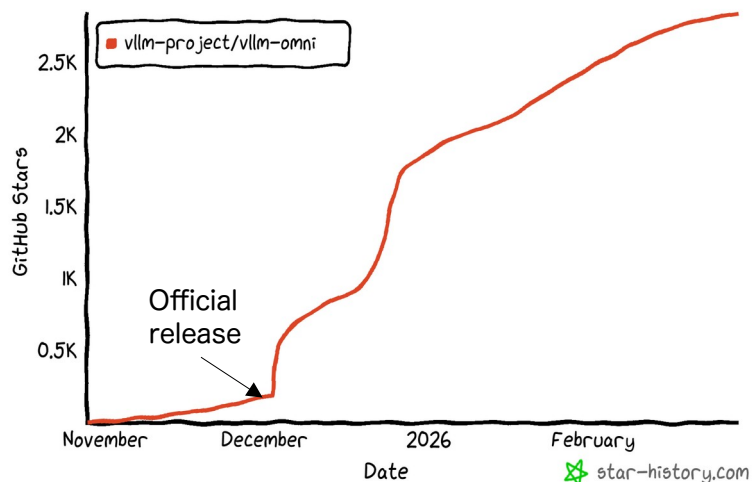
```
$ uv pip install vllm==0.16.0 --torch-backend=auto
```

```
$ uv pip install vllm-omni
```



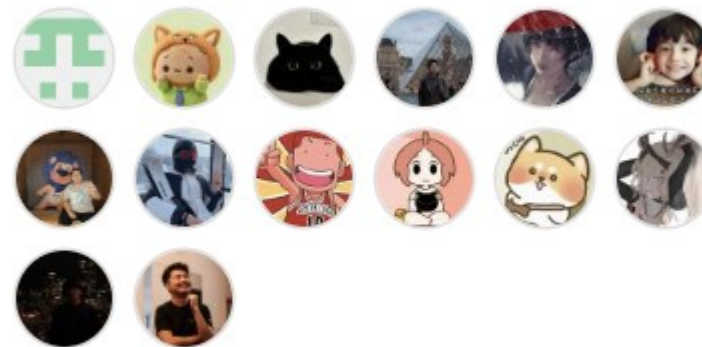
3200+ Stars

Star History



Thanks to all the contributors who raised issues, participated in discussions, and submitted PRs!

Contributors 127



+ 113 contributors



Part 02

Walkthrough

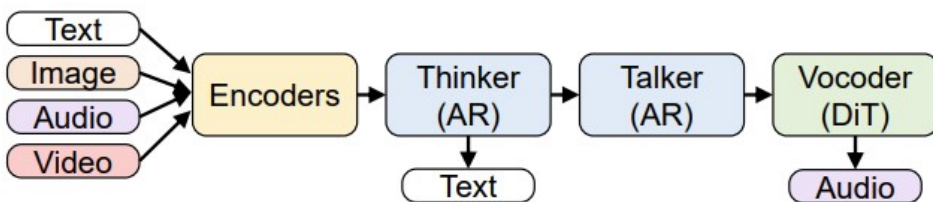


Goal of the walkthrough

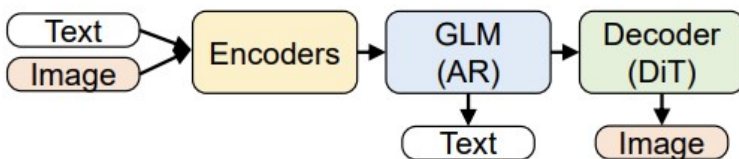
- Understand how vLLM-Omni processes a **multi-modal** request and generates its **multi-modal** outputs.
- Learn where to modify if you would like to make a specific modification/contribution.



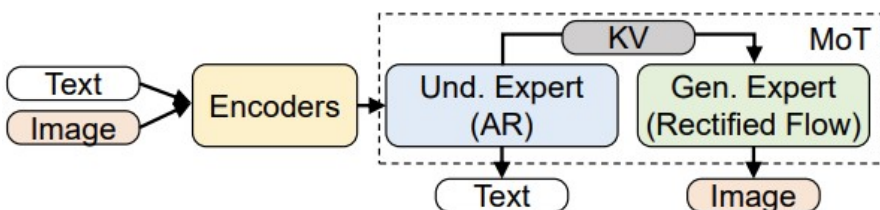
Omni-Modality model classification



(a) Thinker-Talker (Qwen2.5-Omni)



(b) AR + DiT (GLM-Image)



(c) AR + Specialized Generator (BAGEL)

Backbone: (multi) AR + DiT
Models: Qwen-Omni/Ming-Omni
Tasks: any-to-any

Backbone: AR + DiT
Models: Qwen-Image/GLM-Image
Tasks: t2i, t2v, i2i...

Backbone: AR + Spec. Gen.
Models: BAGEL, Hunyuan Image 3.0
Tasks: t2i, i2i, i2t...

Yin, Peiqi, et al. "vLLM-Omni: Fully Disaggregated Serving for Any-to-Any Multimodal Models." arXiv preprint arXiv:2602.02204 (2026).



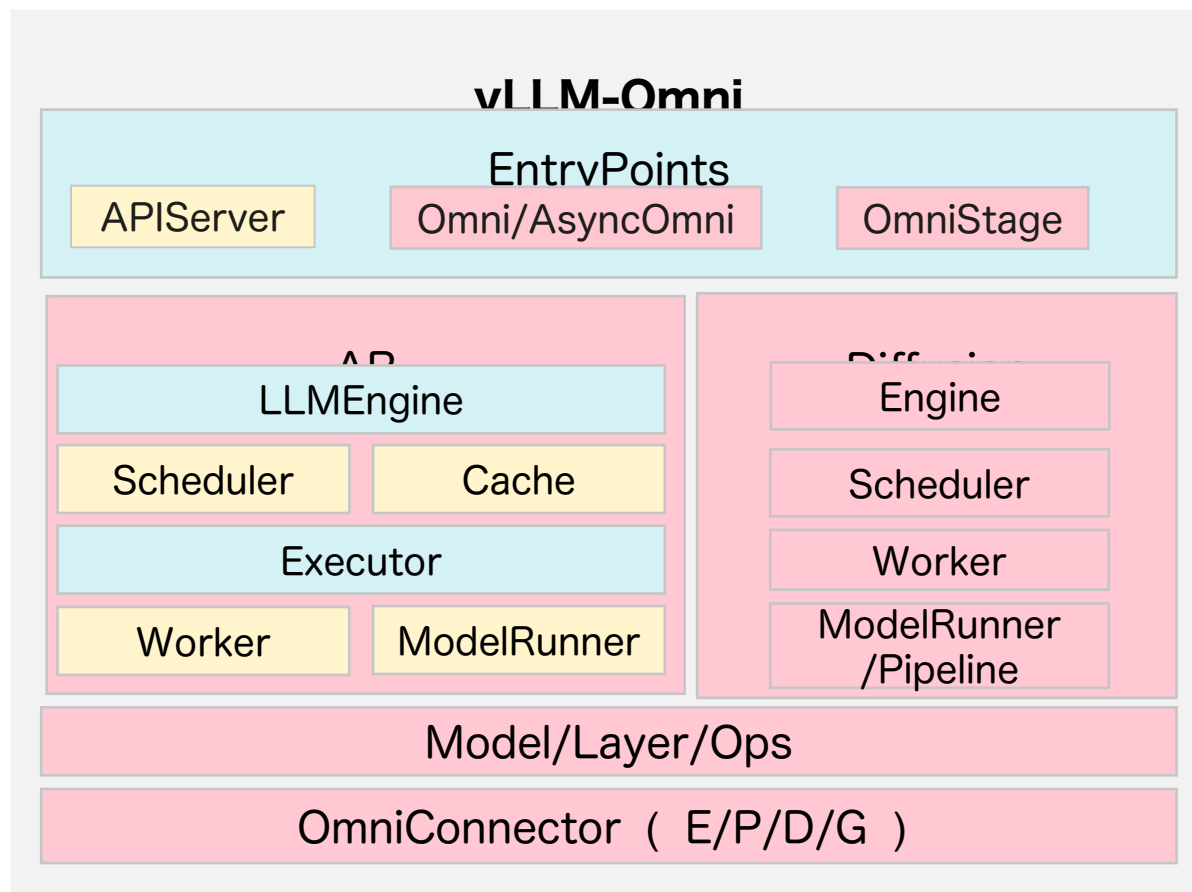
AR/DiT comparison

	AR	DiT
Use cases	Text generation	Multi-modelity generation
Generation process	Token-by-token KV Cache based	Diffusion step
Bottleneck	Prefill: compute bound Decode: memory bound	Compute bound
Seq length	varied	Fixed
Attention Mask	causal mask	Full mask
parallelism	TP/DP/EP/PP/CP/SP	TP/EP/USP/CFG

Yin, Peiqi, et al. "vLLM-Omni: Fully Disaggregated Serving for Any-to-Any Multimodal Models." arXiv preprint arXiv:2602.02204 (2026).



Main architecture of vLLM-Omni



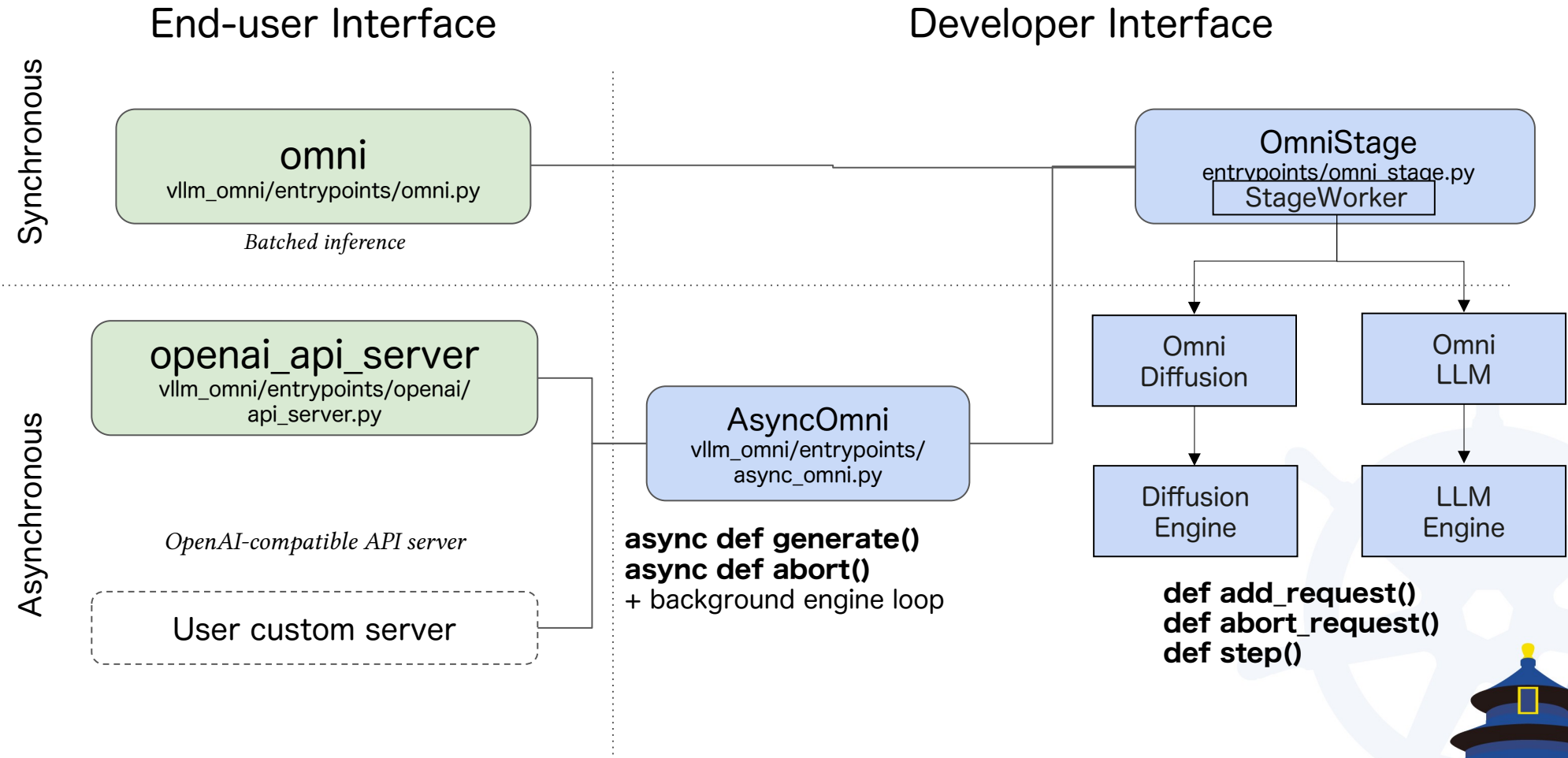
Main component

- **Entrypoints:** offline/online serving, **OmniStage** abstraction for model stages(AR/DiT)
- **AR module:** inherited from vLLM(CB/PA/Prefix Cache...) and adapted to the Omni-modality model
- **Diffusion module:** implemented natively and optimized by acceleration components
- **Model/Layer/ops:** parallelism, quantization, attention
- **OmniConnector:** natively supports E/P/D/G disaggregation

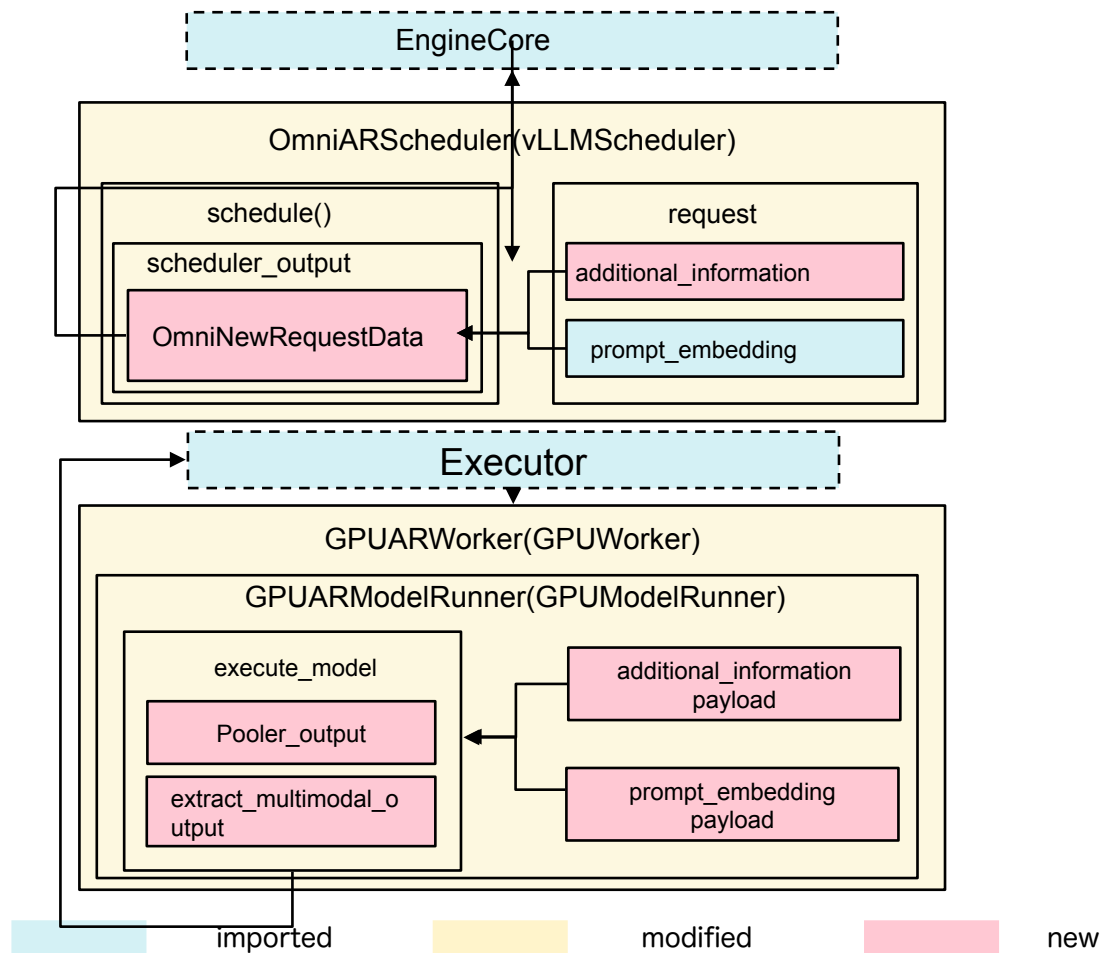
imported modified new



Interface Design



AutoRegressive(AR) Module Design



1. Handles autoregressive generation stages:

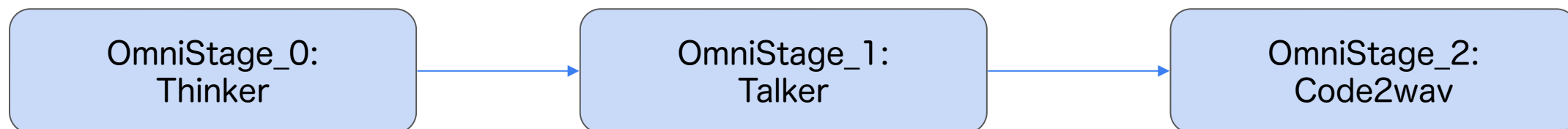
- Text generation
- Hidden embeddings
- multimodal latent tokens (e.g.: Talker in Qwen3-Omni)

2. Extends vLLM's core to support:

- Multimodal inputs/outputs: Processing images, videos, and audio alongside text
- Prompt Embedding: Passing pre-computed prompt embeddings between pipeline stages via serialized payloads
- Additional information: Carrying per-request metadata (tensors, lists) through the pipeline



Config yaml for Qwen3-Omni



stage_id: 0
runtime:
 devices: "0"
engine_args:
 model_stage: thinker
 model_arch: ...ModelClass
 worker_type: ar
 gpu_memory_utilization: 0.9
 engine_output_type: latent
 final_output: true
 final_output_type: text

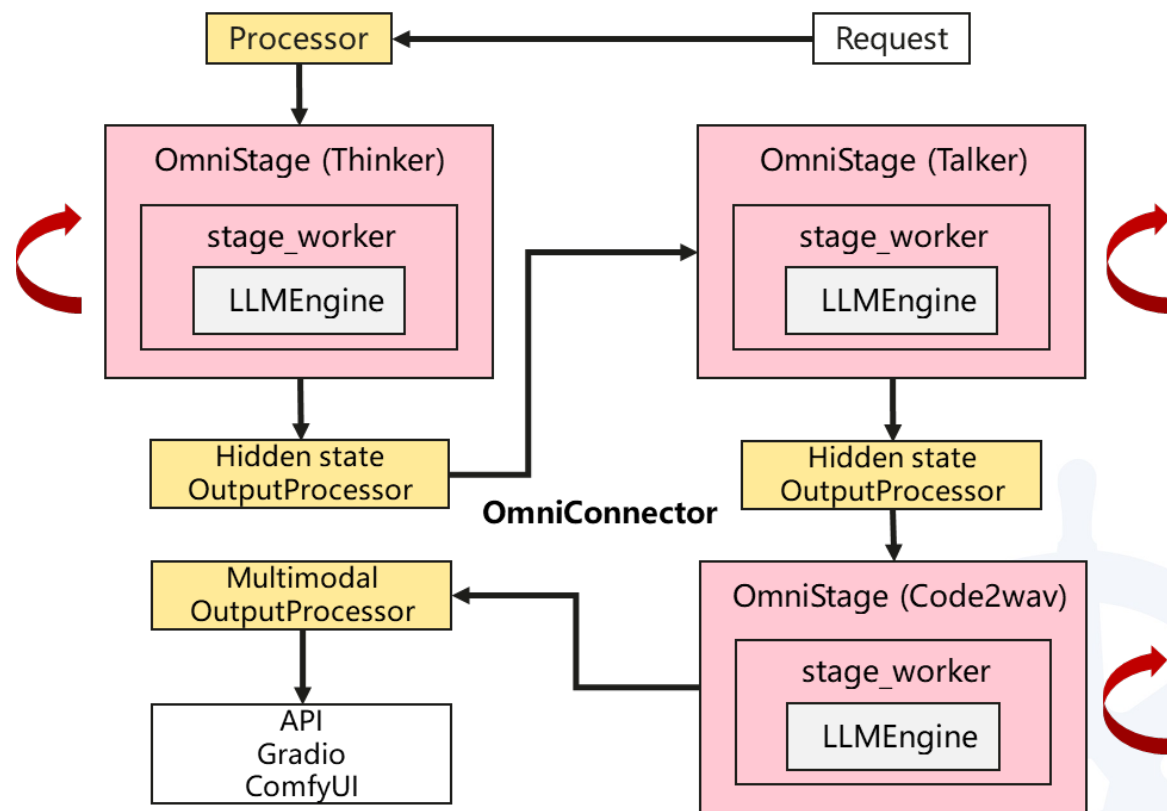
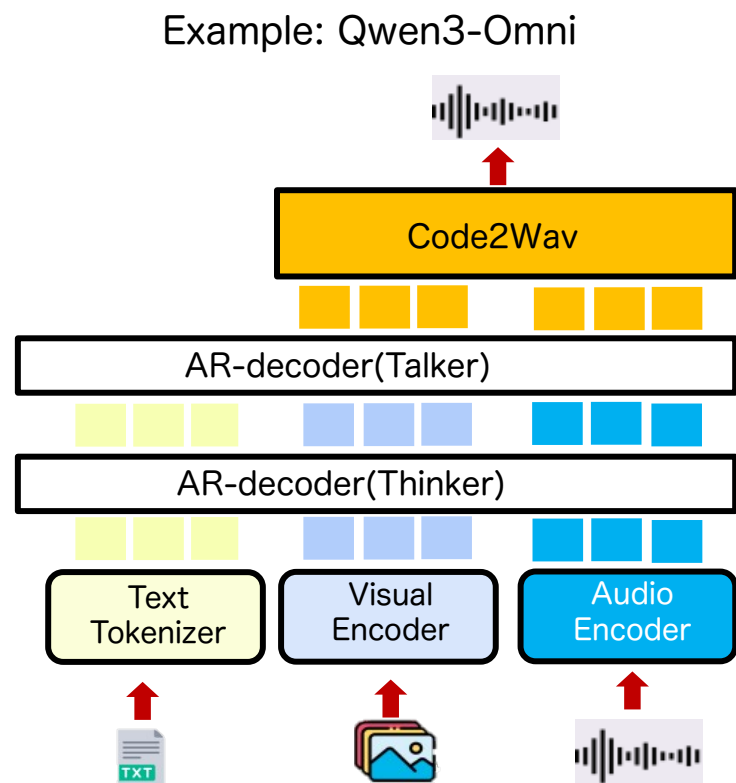
stage_id: 1
runtime:
 devices: "1"
engine_args:
 model_stage: talker
 model_arch: ...ModelClass
 gpu_memory_utilization: 0.6
 engine_output_type: latent
 engine_input_source: [0]

stage_id: 2
runtime:
 devices: "1"
engine_args:
 model_stage: code2wav
 model_arch: ...ModelClass
 gpu_memory_utilization: 0.1

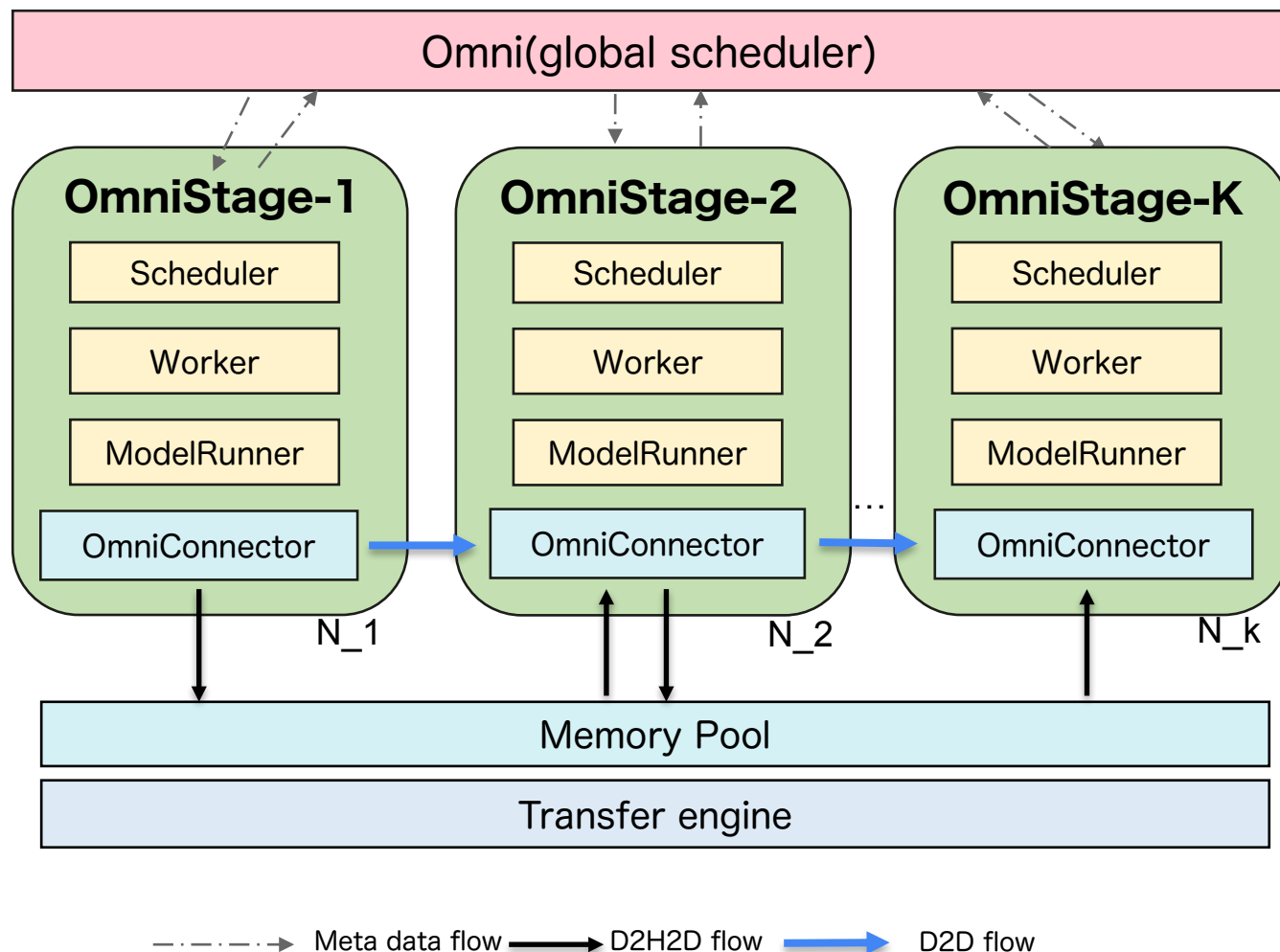
 engine_input_source: [1]
 final_output: true
 final_output_type: audio



When requests arrive



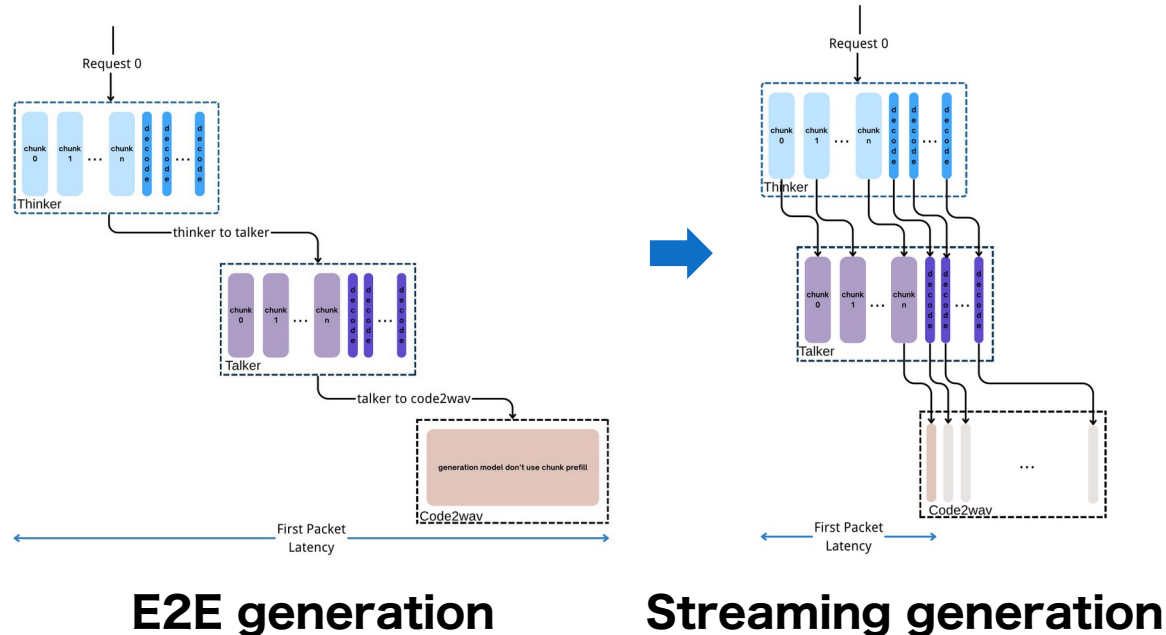
Natively Disaggregated Serving



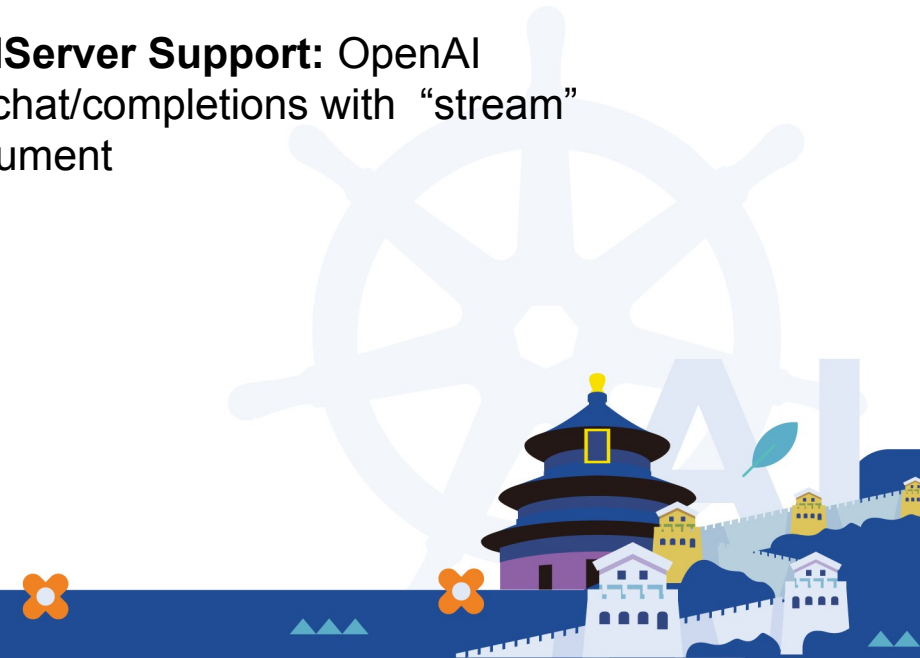
- **Standardized Unified Abstraction:** A generalized interface that handles heterogeneous data (Text, Image, Audio).
- **Control & Data Plane Decoupling:** Metadata travels via lightweight control signals, while heavy payloads are offloaded to high-performance data planes.
- **Hybrid Backend Support:** Native support for Shared Memory (SHM) and Mooncake for distributed transfers.
- **Disaggregated Multi-Modal Execution:** Enables seamless communication between decoupled stages.
- **Multi-Instance Scaling:** Supports multiple instances for each omni stage, enabling elastic deployment and efficient load distribution across distributed clusters.



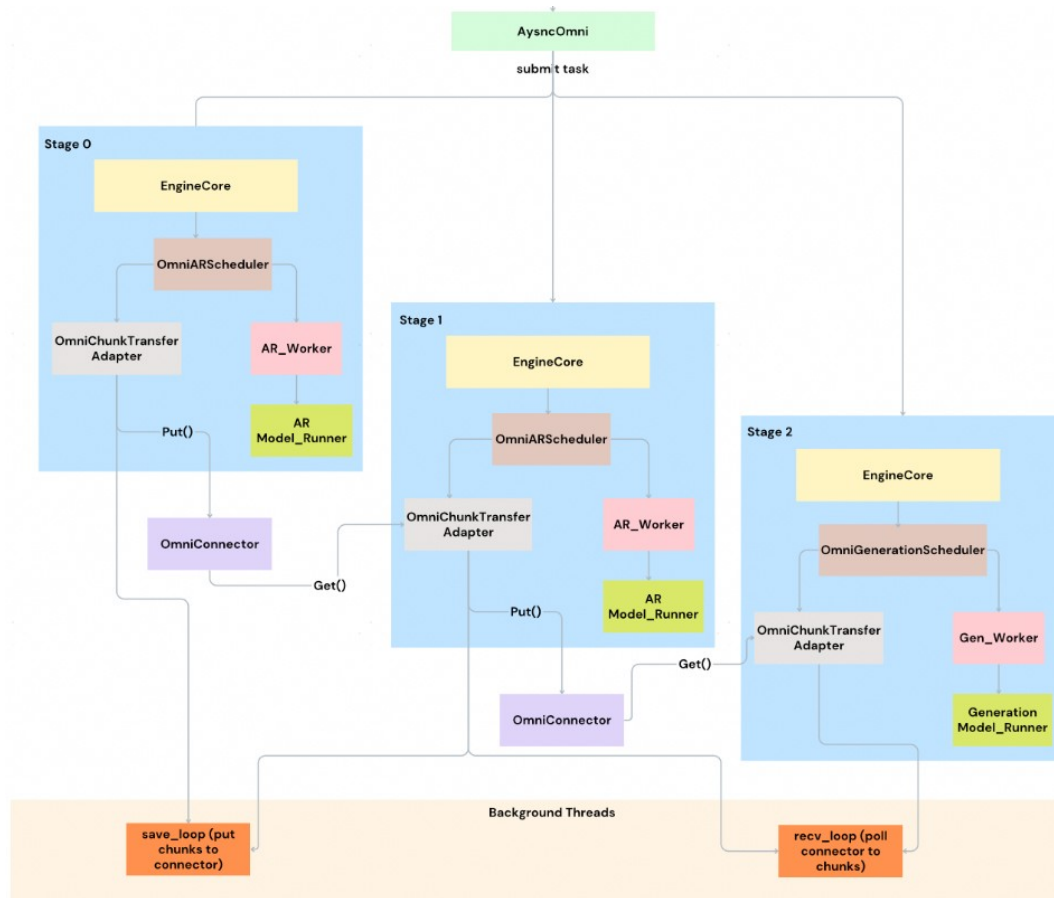
Async Chunk & Audio Streaming Output



- **Pipeline Between Stages:** Asynchronous chunked computation and communication across stages
- **Audio Streaming Output:** The waveform is output immediately after Talker generates each token.
- **API Server Support:** OpenAI v1/chat/completions with “stream” argument

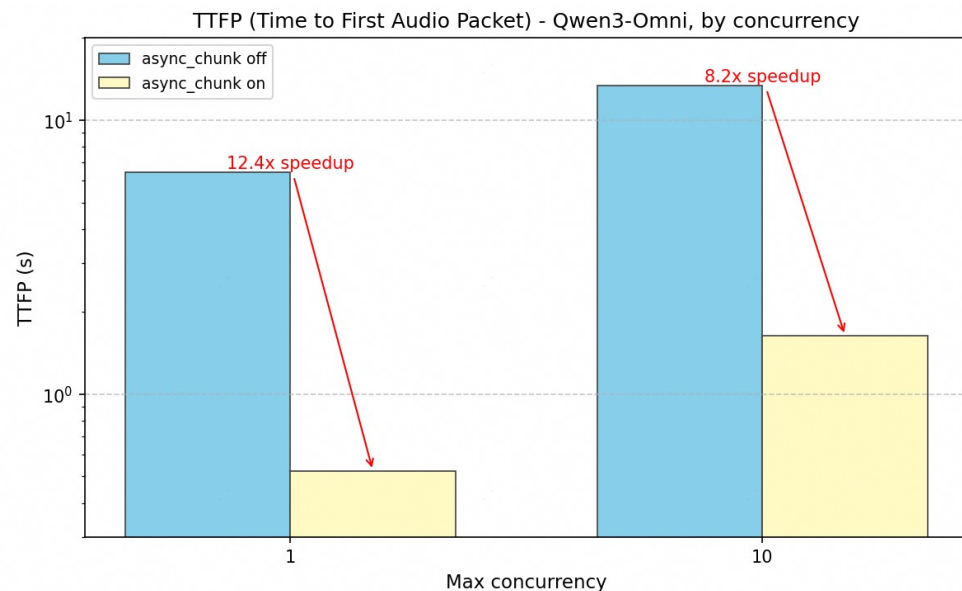


Async Chunk & Audio Streaming Output

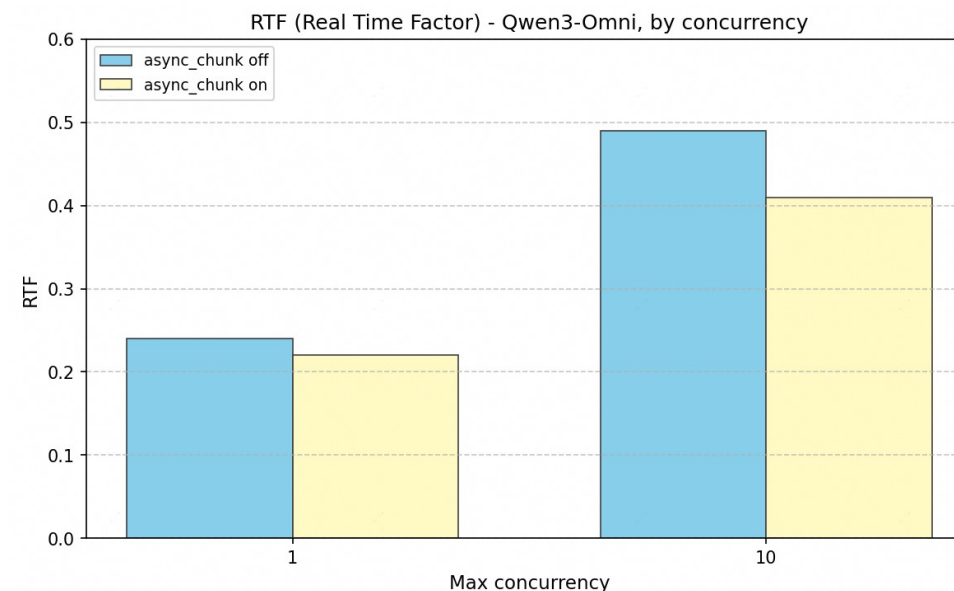


- **OmniConnector:** Transmit data between stages.
- **OmniChunkTransferAdapter:** Chunk-specific implementation that owns the full chunk lifecycle when `async_chunk` is enabled.
- **Stage Input Processors:** Custom functions that process stage outputs into chunks for different models.
- **Schedulers:** Modified to handle chunk-based scheduling with async IO-compute overlap.
- **Model Runners:** Handle chunk processing.

Qwen3-Omni: 8~12 TTFP reduction



1 / 10 concurrency: 12.4x / 8.2x improves



1 / 10 concurrency: 1.1 x / 1.2 x improves

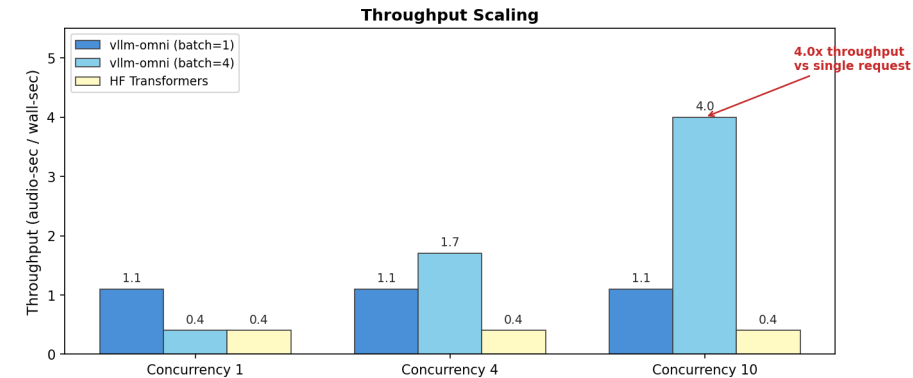
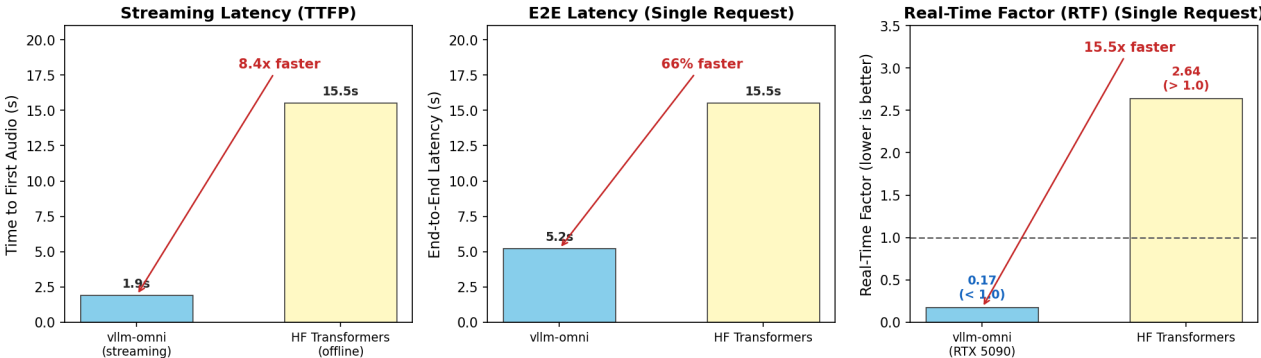


Qwen3-TTS: RTF 15x improvement



Qwen3-TTS 1.7B: vllm-omni vs HuggingFace Transformers

Model: Qwen3-TTS-12Hz-1.7B-CustomVoice | GPU: NVIDIA H200 NVL | 50 requests per test
Without specification, H200 is used for testing.



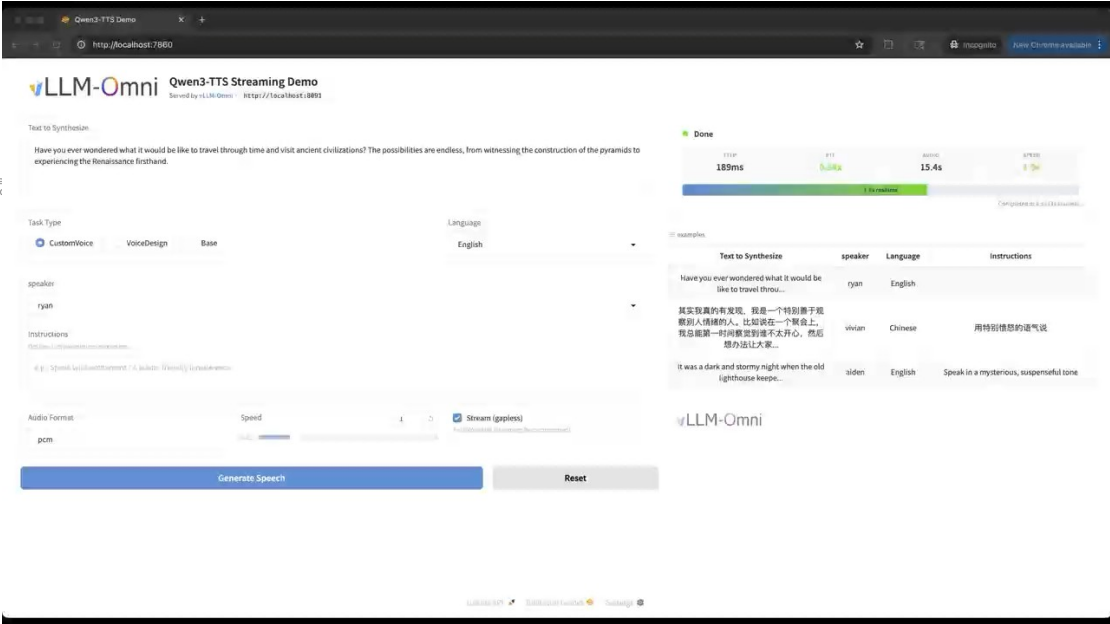
Key Improvements

8.4x
faster time-to-first-audio
vs HF (streaming)

66%
lower E2E latency
at single concurrency

RTF 0.17
real-time capable
(HF: 2.64)

4.0x
throughput scaling
with batch=4



Omni-Modality models Acceleration Support



LLM-Omni

Model	Output Modalities	Disaggregated deployment	Streaming	Async chunk	Graph	Gradio demo
Qwen2.5-Omni	Text/Audio	✓	Only text	✗	✗	✓
Qwen3-Omni	Text/Audio	✓	✓	✓	✓	✓
MiMo-Audio	Text/Audio	✓	✓	✓	✓	✗
Qwen3-TTS	Audio	✓	✓	✓	✓	⌚
Bagel	Text/Image	✓	Only text	N/A	Only AR	✓
GLM-Image	Image	✓	N/A	N/A	✓	⌚
Hunyuan-Image	Text/Image	✓	Only text	N/A	Only AR	✓

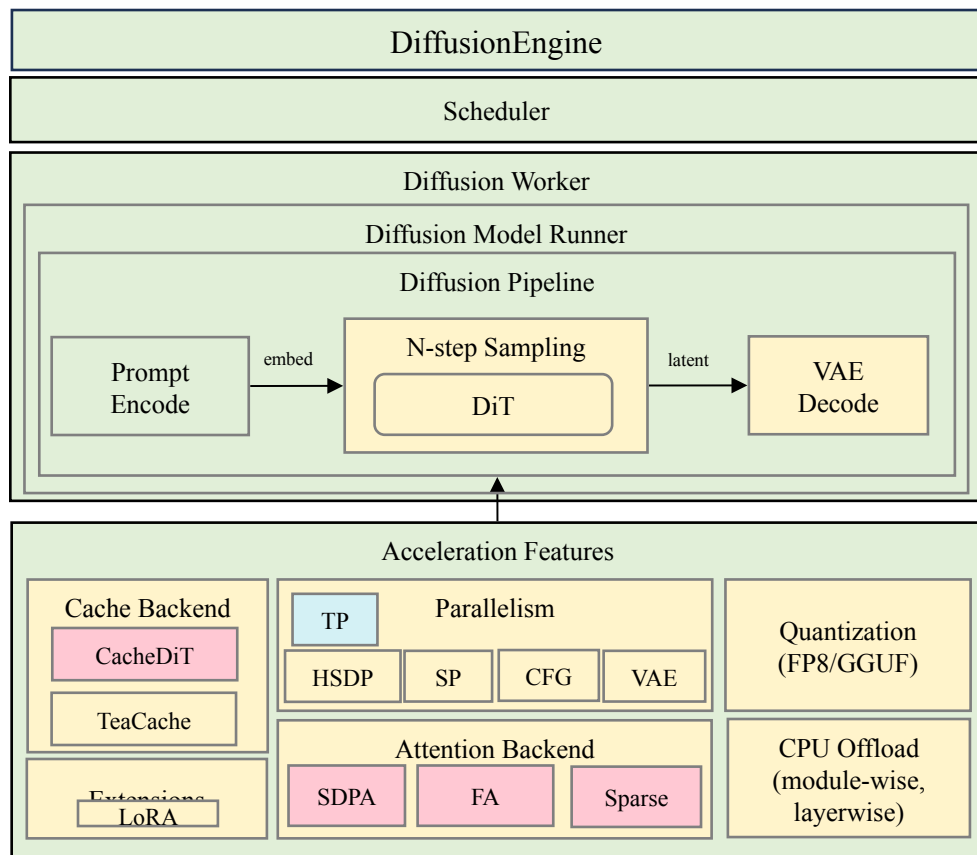
✓ = supported

✗ = not yet supported

⌚ = in progress (PR under review)



Diffusion Module Design



Legend: ■ vllm import ■ native ■ Third-party

Diffusion Core

- Natively implemented and optimized by configuring the acceleration layer
- Aligned with the AR arch with model_runner abstraction
- *Encoders are disaggregated to run on vllm engine for higher throughput in large-scale deployment*
- *Support step-wise scheduler with continuous*
- *Connect with AR module to speed up AR+DiT model inference with OmniConnector*

Acceleration Features

- **Cache backend:** Cache-DiT, TeaCache...
- **Parallelism:** TP/HSDP/SP/CFG/VAE...
- **Attention:** interface abstraction for third-party integration(FA/SAGE/Sparse...)
- **Quantization:** FP8/GGUF...
- **CPU Offload:** module-wise/layer-wise
- **Extensions:** LoRA



Diffusion modules with Acceleration Support

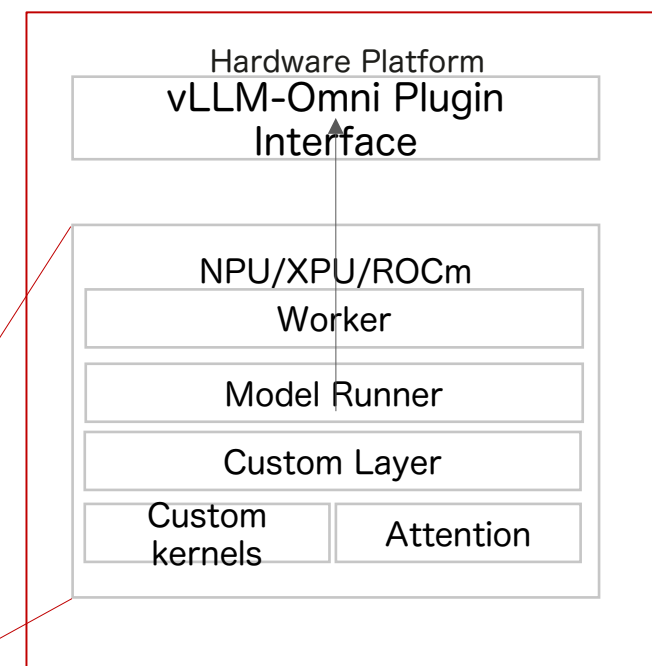
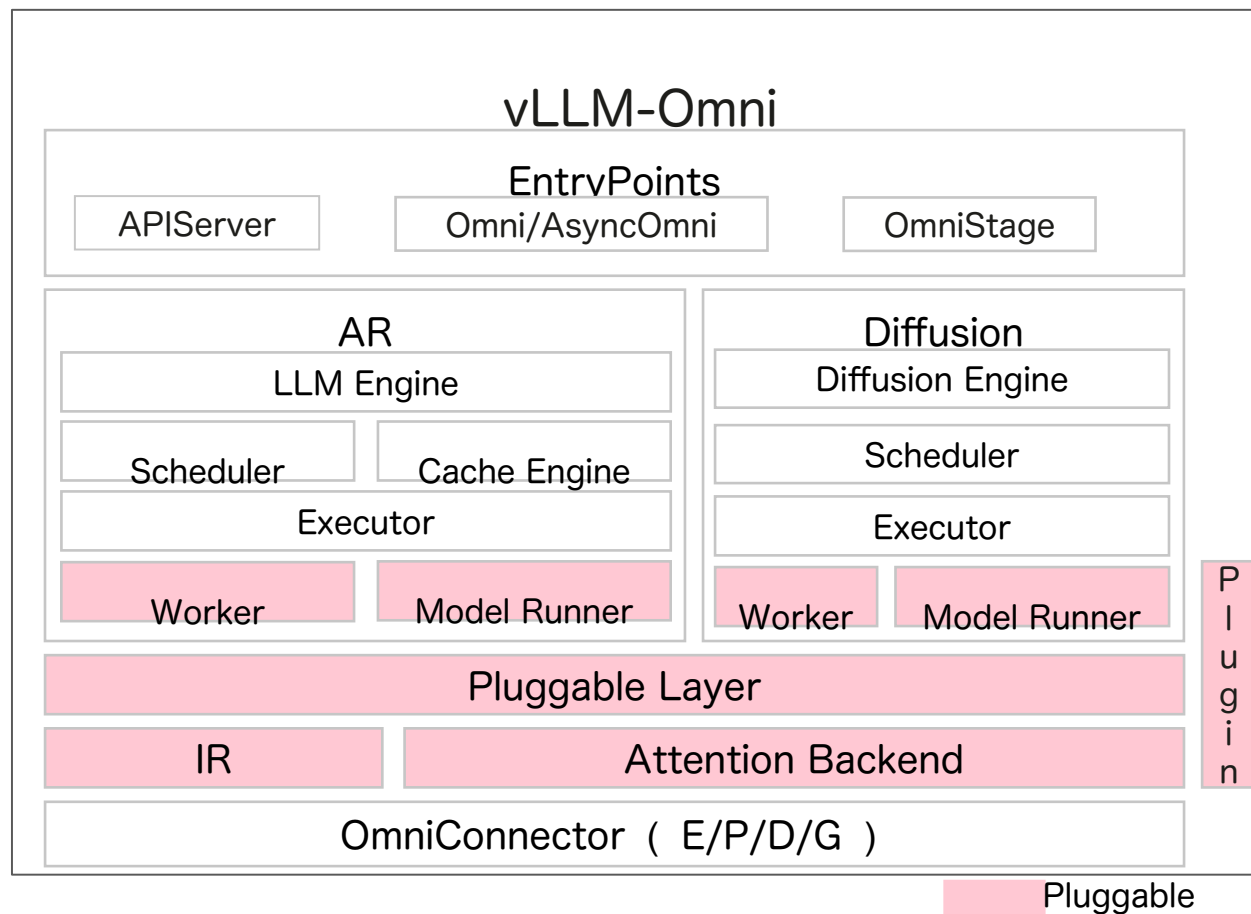
Model	Output modality	TeaCache	Cache-DiT	SP (Ulysses & Ring)	CFG-Parallel	TP	HSDP	VAE-Patch-Parallel	CPU Offload (Layerwise)	Quant (fp8/gguf...)
FLUX.1-dev	Image	⌚	✓	✗	✓	✓	✗	✗	✗	✓
FLUX.2-klein		⌚	⌚	⌚	✓	✓	✗	✗	✗	✓
LongCat-Image		✗	✓	✓	✓	✓	✗	✗	✗	✗
LongCat-Image-Edit		✗	✓	✓	✓	✓	✗	✗	✗	✗
Ovis-Image		✗	✓	✗	✓	⌚	✗	✗	✗	✗
Qwen-Image		✓	✓	✓	✓	✓	✓	✗	✓	✓
Qwen-Image-Edit		✓	✓	✓	✓	✓	✓	✗	✓	✓
Qwen-Image-Layered		✗	✓	✓	✓	✓	✗	✗	✓	✗
SD3.5		⌚	✓	⌚	✓	✓	✗	✗	✗	✗
NextStep1.1		⌚	⌚	✗	✗	⌚	✗	✗	✗	✗
Z-Image		✓	✓	✓	✗	✓	✗	✓	✗	✓
Wan2.2	Video	✗	✓	✓	✓	✓	✓	✓	✓	✓
HunyuanVideo1.5		⌚	✓	✓	✓	✓	✓	✗	⌚	⌚
Seed-Helios		✗	✗	✗	✗	⌚	✗	✗	✗	⌚
Stable-Audio-Open	Audio	⌚	⌚	✗	✗	⌚	✗	✗	✗	✗

• ✓ = supported

• ✗ = not yet supported

• ⌚ = in progress (PR under review)

Hardware Plugin System



Summary



- 1. Support Omni-Modality model inference & serving
- 1. Consistent and unified API interface with vLLM
- 1. Native disaggregated deployment for different model stages
- 1. Pipeline async streaming for multiple stage engines
- 1. Native support for diffusion stage acceleration



Part 03

Release roadmap



Latest Release and Schedule



vLLM-Omni v0.16.0

 **Qwen3-Omni TTFL drops 90%**

121 Commits · ~60 Contributors (24 New )

-  **Ultra-Fast Audio & Video Engine**
 -  Qwen3-Omni / TTS Performance Surge
TTFL ↓ 90% · RTF 0.22~45
 -  Production-Grade MiMo-Audio
Speed up 11x faster · RTF ≈0.2
 -  Major Video & Image Generation (DIT) Upgrade
★ OPENAI-Compatible Video API
 -  DIT Extreme Acceleration & Quantization
FP8/GGUF · TP/HSDP · SD 3.5
-  **New Models & Multimodal Expansion**
 -  Video / Image: HunyuanImage3, OmniGen2, nextstep_1
 -  ComfyUI Integration
 -  RDMA High-End Deployment
Full Hardware: Intel, AMD, NPU

- **RC version**
Inherited new features from latest vLLM
Matched with vLLM formal release (e.g. v0.16.0rc1 released on Feb 13, 2026 for vLLM v0.16.0)
- **Stable version**
New features with sufficient tests
Every 4 weeks (e.g. v0.16.0 released on Feb 28, 2026)



Community collaboration

- Day-0 Support for SOTA models
- Hardware support (Cuda/ROCm/XPU/NPU)
- RL integration with veRL&veOmni



Future Roadmap (Overview)

P0:

- Stabilize our core API and user interface
- CI/CD
- Hardware Support(CPU/GPU/xPU...)
- Basic optimization (Async chunk/Parallel/Cache)
- Better benchmarks

P1:

- Wider model support(Omni&DiT...)
- Continuous performance optimization(SparseAttn, CPU offloading, prefix cache, Quantization, Diffusion Continuous Batching)
- Full disaggregation(OmniConnector&OmniRouter...)
- Large scale deployment
- RL Integration



Part 04

Connect us



vLLM Networking Hour!



<https://github.com/vllm-project/vllm-omni>



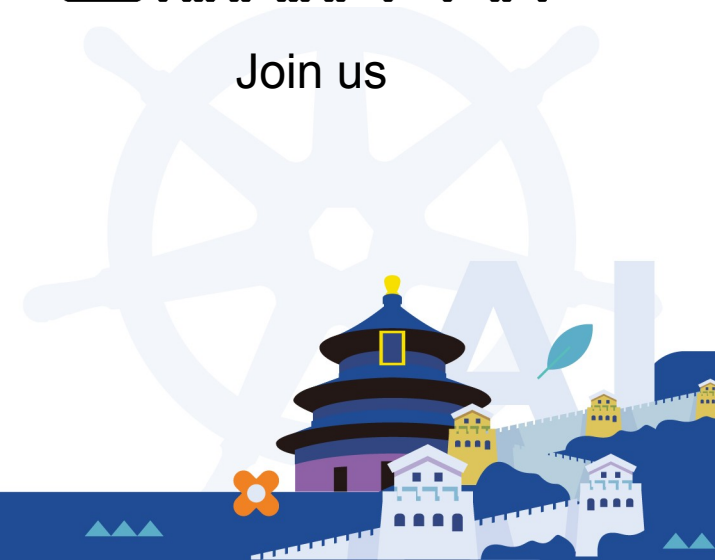
<https://blog.vllm.ai/2025/11/30/vllm-omni.html>



<https://communityinviter.com/apps/vllm-dev/join-vllm-developers-slack>



Join us



Thanks.

